Early Evidence on Generative AI Exposure and Labor Market Outcomes

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Github Repository

Stakeholders

This memo is written for a Director of Workforce Policy at a state Department of Labor. The intended audience is assumed to be comfortable interpreting applied research and labor market indicators, but not interested in technical implementation details. The priority is actionable evidence that can inform workforce planning and early-career employment policy.

Executive Summary

This memo examines whether the spread of generative AI is associated with early changes in labor market outcomes, and whether those changes appear to fall more heavily on younger workers. Using CPS ASEC data from 2016-2025, the analysis links workers to an industry-level AI exposure measure (AIIE) and studies how employment conditions shifted once the GenAI period began.

Overall, employment conditions worsened somewhat more in highly AI-exposed industries after 2023, particularly for workers ages 25–34. The evidence is much weaker for wages, where no parallel signal emerges over the short post period now available.

These results should be interpreted cautiously. The unemployment analysis is more informative than the wage analysis, but neither should be treated as definitive causal proof. Even so, the evidence is strong enough to justify closer monitoring of younger workers in AI-exposed sectors and to treat emerging employment frictions in those sectors as a policy-relevant risk.

Decisions To Be Made

- Whether to prioritize monitoring and support for younger workers in AI-exposed sectors, especially workers in the 25-34 range.
- Whether to expand labor-market monitoring so that hiring, job finding, and layoffs can be separated more clearly in future work.
- Whether to treat wage changes as a secondary indicator and focus policy attention first on employment outcomes.

The Problem

Public debate about generative AI often centers on job displacement or wage pressure. The more immediate question here is narrower and more practical: are early signs of strain already appearing in the parts of the labor market that are most exposed to AI, and if so, which workers appear most affected?

That framing matters because this analysis covers workers ages 25-34 as well as older workers with more established careers. The key question is whether employment conditions appear to be weakening faster for earlier-career workers in AI-exposed industries than for older workers. The Current Population Survey Annual Social and Economic Supplement has limited data on layoff identification (for example, distinguishing layoffs from quits), so the results are best interpreted as evidence on labor market outcomes rather than on a single causal source.

Analytical Approach

To answer that question, CPS ASEC data from 2016-2025 were merged with an industry-level AI exposure measure, AIIE. The analysis then compares how unemployment and wages shifted after the GenAI period began, contrasting industries with higher AI exposure against those with lower exposure.

The two outcomes are unemployment and wage income. The analysis also looks separately at younger workers ages 25-34 to assess whether labor market changes appear more concentrated earlier in workers' careers. Examples of relatively high-AI-exposure industries in this sample include legal services, accounting and payroll services, computer systems design, and management consulting. Lower-exposure examples include landscaping,services to buildings and dwellings, warehousing and storage, and truck transportation.

The main empirical setup is a difference-in-differences comparison. In plain terms, the method asks whether outcomes (unemployment and wage) changed more after the GenAI period began in industries with higher AI exposure than in industries with lower AI exposure, after accounting for basic worker characteristics. This helps separate a broad, economy-wide shift from a change concentrated specifically in AI-exposed industries.

$$Outcome_{it} = \alpha + \beta_1 AIIE_j + \beta_2 Post_t + \beta_3 (AIIE_j \times Post_t) + Controls + \epsilon_{it}$$

For unemployment, the post period begins in 2023. For wages, the post period begins in 2024 because 2023 is treated as a transition year. The models use CPS ASEC survey weights, control for age and education, and cluster standard errors at the industry level. Additional checks examine whether the results remain similar when 2020 is excluded and whether the pattern differs across age groups. The goal is not to claim perfect identification, but to see whether the same basic signal holds across several reasonable ways of looking at the data.

Results At A Glance

The table below summarizes the main regression signals and the most relevant diagnostic check, as a reference point before looking at the figures and more in depth narrative interpretations.

Result	Estimate	P-Value	Interpretation
Main unemployment DiD	0.0029	0.0143	Higher AI exposure is associated with a modest relative increase in unemployment after 2023.
Younger-worker unemployment effect	0.0034	0.0099	The unemployment shift is larger for workers ages 25-34 than for older workers.
Main wage DiD	0.0026	0.7297	No meaningful wage response is detected in the short post period.
Unemployment pre-trend check	Joint test	0.0021	Prep-period differences remain, so the unemployment result is informative but should still be treated cautiously
Wage pre-trend check	Joint test	0.0000	The wage comparison is less convincing because pre-period differences are stronger.
Age 25-29 vs 30-34 gap	0.0022	0.4210	The younger subgroup estimate is larger, but the two age subgroups are not statistically distinct.

Descriptive Patterns

The evidence develops in stages. The first step is to examine the broad patterns in the data before turning to the regression results. High-AI-exposure industries generally have lower unemployment than low-exposure industries throughout the sample, but that gap narrows after the GenAI period begins. That does not necessarily mean high-exposure industries suddenly become high-unemployment industries, but instead, it suggests that their pre-existing advantage becomes smaller in the post period.

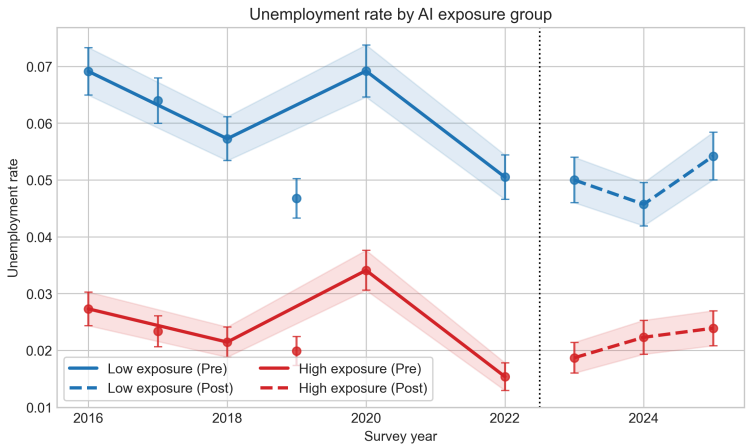

Figure 1. Higher-AI-exposure industries maintain lower unemployment overall, but that advantage narrows after 2023. Annual unemployment averages are shown with separate pre/post LOWESS fits and 95% confidence bands.

The age-focused descriptive plots point in the same direction. The relative post-period unemployment shift appears more pronounced for workers ages 25-34 than for older workers, although the descriptive evidence alone does not identify the reasoning behind that difference. At this stage, the figures are best read as directional evidence that helps motivate the more formal comparisons that follow.

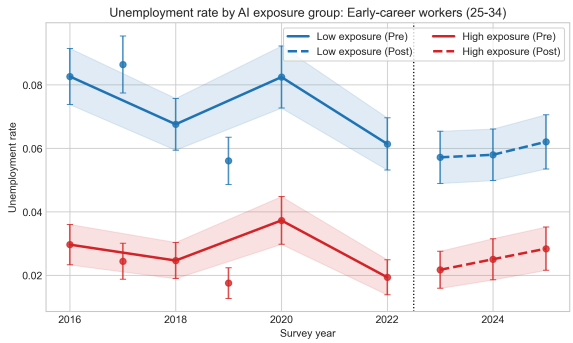

Figure 2. The post-2023 unemployment shift appears more pronounced for workers ages 25–34. Annual unemployment averages are shown with separate pre/post LOWESS fits and 95% confidence bands.

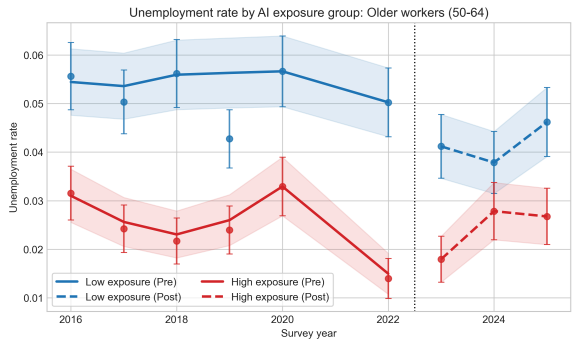

Figure 3. For workers ages 50–64, the same post-period narrowing is less pronounced. Annual unemployment averages are shown with separate pre/post LOWESS fits and 95% confidence bands.

The older-worker comparison provides useful context. For workers ages 50–64, the same pattern is present much less strongly, which is consistent with the later regression evidence that the post-period shift is concentrated more heavily among earlier-career workers.

The wage plot is included to the right, but it shows much less visual separation by exposure group after the GenAI period begins, which is why we focus our memo more on the unemployment results.

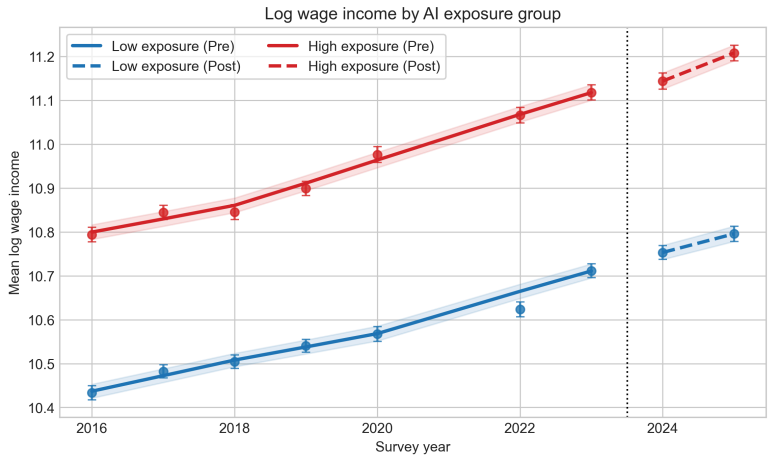

Figure 4. Wage patterns show much less post-period separation by AI exposure group. Annual log wage averages are shown with separate pre/post LOWESS fits and 95% confidence bands.

Main Findings

Taken together, the analysis points to four findings that are most relevant for near-term labor market monitoring and policy planning.

Higher AI exposure is linked to a modest post-2023 increase in unemployment

The main unemployment model indicates that the post-2023 change is less favorable in industries with higher AI exposure than in industries with lower exposure. The magnitude is modest, not severe, and it does not imply that high-exposure industries become the weakest parts of the labor market overall. Instead, what changes is that their earlier advantage becomes smaller in the post period. That shift is meaningful because it may be one of the first measurable signs of strain in the sectors most exposed to AI.

The unemployment shift is more pronounced for workers ages 25-34

This matters because the labor market shift does not appear to be evenly distributed across workers. Once the model allows the post-period effect to vary by age, the deterioration is meaningfully larger for workers ages 25–34 than for workers ages 35–64, even after accounting for age, education, and industry

clustering. The most defensible interpretation is that early-career workers in more AI-exposed industries are facing weaker employment conditions than older workers in the same period.

The youngest subgroup has the largest point estimate, but the split within 25-34 is not conclusive
Breaking the younger cohort into narrower age bands suggests that the estimated effect is largest for workers ages 25–29. However, the gap between the 25–29 and 30–34 groups is not estimated precisely enough to justify a firm claim that the youngest subgroup is uniquely affected. The safest reading is that the vulnerability appears concentrated somewhere within the broader early-career range, without strong evidence of a sharp dividing line inside that group.

No meaningful wage effect is visible in the short post period
The wage models do not show a meaningful divergence after 2024 between high- and low-exposure industries. This should not be taken as evidence that AI has no effect on wages. Rather, it suggests that any wage response is either small, delayed, or difficult to distinguish from pre-existing differences over the short window currently available. At this stage, wages are better treated as a secondary indicator than as the main basis for action.

Why More Weight Is Placed On Unemployment

A finding is only useful if we are clear about how much confidence it can support. To interpret the post-period differences credibly, it is important to check (parallel trends) whether higher- and lower-AI-exposure industries were already moving apart before generative AI became widespread. If they were already diverging before 2023, it is harder to attribute later differences to the AI period itself. If they were moving more similarly before 2023 and only separate afterward, the unemployment result becomes more persuasive. This is a model check rather than a separate finding, but it matters for judging how much confidence to place in the results.

This diagnostic is more reassuring for unemployment than for wages. For unemployment, the pre-period differences are smaller and less systematic. For wages, higher-AI-exposure industries had different levels and trajectories well before the post period, which makes the wage comparison less convincing as evidence of a distinct AI-related shift. That is the main reason the memo places more emphasis on unemployment than on wage changes.

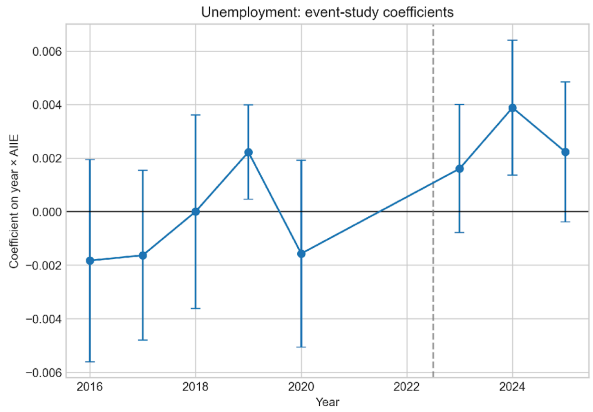

Figure 5. Pre-period differences are more manageable for unemployment than for wages, which is why more weight is placed on the unemployment result. The figure shows year-by-year comparisons using 2022 as the reference year.

Robustness and Limitations

A robustness check excluding 2020 reaches the same conclusion. The younger-worker unemployment pattern remains visible after removing the most unusual COVID year, which reduces concern that the main result is being driven entirely by pandemic-era volatility. At the same time, these checks improve confidence only at the margin; they do not turn the analysis into definitive causal proof. The practical takeaway is that the core unemployment result appears reasonably stable, but it should still be used as an input into policy judgment rather than as a stand-alone basis for major action.

Several limitations should shape how the results are used. First, the CPS is a repeated cross-section rather than a panel, so the same worker cannot be followed over time. Second, AIIE measures potential AI exposure based on industry task content, not actual AI adoption by a specific employer. Third, the data do not distinguish cleanly between reduced hiring, weaker job-finding, and layoffs, so the unemployment results do not identify a single mechanism. Fourth, the post period is still short, especially for wages. Finally, the supporting diagnostic checks are more favorable for unemployment than for wages, but they are not strong enough to remove all concern about pre-existing differences across industries.

Conclusion

The story that emerges from the analysis is fairly consistent. Labor market conditions appear to have weakened somewhat more after 2023 for younger workers in higher-AI-exposure industries. The unemployment comparison points in that direction, and the age breakdown suggests the pattern is stronger for workers ages 25-34 than for older workers.

At the same time, the memo should not be read as definitive proof that generative AI caused these changes. The findings are better understood as an early warning signal than as a final causal verdict. The most practical implication is that younger workers in AI-exposed sectors warrant closer monitoring, while wage effects are not yet clearly visible in the data. If resources for monitoring or intervention are limited, this evidence suggests they are likely to be most useful when directed toward early-career workers in the more AI-exposed parts of the labor market.

Appendix: References

AIOE-Data. n.d. "AIOE and Related Datasets." GitHub repository. Accessed May 1, 2026. https://github.com/AIOE-Data/AIOE.

Felten, Edward, Manav Raj, and Robert Seamans. 2021. "Occupational, Industry, and Geographic Exposure to Artificial Intelligence: A Novel Dataset and Its Potential Uses." Strategic Management Journal 42 (12): 2195-2217. https://doi.org/10.1002/smj.3286.

Flood, Sarah, Miriam King, Renae Rodgers, Steven Ruggles, J. Robert Warren, Daniel Backman, Annie Chen, Grace Cooper, Stephanie Richards, Megan Schouweiler, and Michael Westberry. 2024. Integrated Public Use Microdata Series, Current Population Survey: Version 12.0 [dataset]. Minneapolis, MN: IPUMS. https://doi.org/10.18128/D030.V12.0.

U.S. Census Bureau. n.d. "2017 Census Industry Code List with Crosswalk." Industry and Occupation Code Lists & Crosswalks. Accessed May 1, 2026. https://www.census.gov/topics/employment/industry-occupation/guidance/code-lists.html.

U.S. Census Bureau. n.d. "2018 Occupation Code List and Crosswalk." Industry and Occupation Code Lists & Crosswalks. Accessed May 1, 2026. https://www.census.gov/topics/employment/industry-occupation/guidance/code-lists.html.

U.S. Census Bureau. n.d. "2022 Census Industry Code List with Crosswalk." Industry and Occupation Code Lists & Crosswalks. Accessed May 1, 2026. https://www.census.gov/topics/employment/industry-occupation/guidance/code-lists.html.